

Visualizations on Smart Watches while Running: It Actually Helps!

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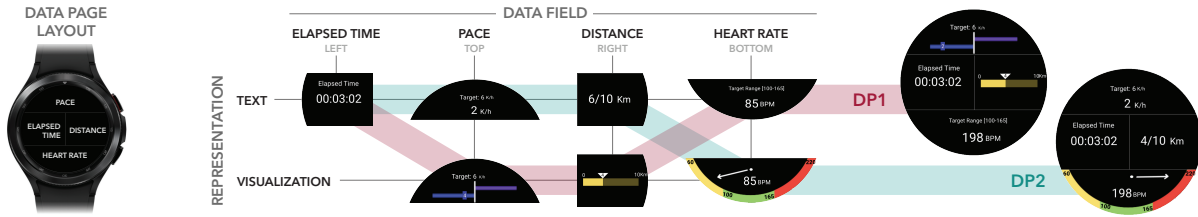


Figure 1: Data Page layout showing Elapsed Time (left), Pace (top), Distance (right) and Heart Rate (bottom). Pace, Distance and Heart Rate are represented with TEXT or VISUALIZATION. DP1 shows Elapsed Time and Heart Rate with TEXT, and Pace and Distance with VISUALIZATION; DP2 shows Elapsed Time, Pace and Distance with TEXT, and Heart Rate with VISUALIZATION.

ABSTRACT

Millions of runners rely on smart watches that display running-related metrics such as pace, heart rate and distance for training and racing — mostly with text and numbers. Although research tells us that visualizations are a good alternative to text on smart watches, we know little about how visualizations can help in realistic running scenarios. We conducted a study in which 20 runners completed running-related tasks on an outdoor track using both text and visualizations. Our results show that runners are 1.5 to 8 times faster in completing those tasks with visualizations than with text, prefer visualizations to text, and would use such visualizations while running — if available on their smart watch.

Index Terms: Running, Visualization, Smartwatch visualization.

1 INTRODUCTION AND BACKGROUND

Running is the most accessible and universal physical activity — after all, endurance running has shaped humans’ physiology and anatomy [5]. Modern runners do not necessarily run for survival anymore, but for benefits in improving body health, competitiveness, or leisure [4, 15, 20] — running is the most-uploaded sport on the Strava sports-focused social network [31]. About two-thirds of runners rely on smart watches while they run [18] to track various running-related **data fields** shown on what are called **data pages**.

Surprisingly though, these data fields are typically represented with text and numbers (e.g., [14, 16, 24, 26, 30]), not visuals [1, 17], despite indications that visual elements [7, 8, 11, 28] and charts [27, 29] can help when performing physical activities. Studying visualizations on smart watches while running is promising because in this scenario, shifting [25] the runner’s attention from their environment (and main task of running) to the display (and the task of reading running metrics) breaks the running flow [19, 22, 32]. Looking at a visualization, even for a few seconds, can also be challenging, such as when on the road with traffic, racing with hundreds of other runners, or on technical treacherous trails. In these contexts, glanceable visualizations are likely to be useful — albeit not straightforward to design given the small screen-size and low resolution of smart watches [3].

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We know that glanceable feedback [13] on smart watches can help for everyday activity tracking of health and activity levels; that it is possible to retrieve information from glanceable visualizations on smart watches [3]; and that real-time visualizations can be useful for runners while running on a treadmill — an indoor, low-risk environment [29]. However, we do not know whether this holds true in realistic running settings [1] wherein runners might experience reduced cognition [9] and must focus more on their environment than on their wearable devices [19, 25].

We contribute to filling this gap with a study in which runners performed *realistic* tasks in a *realistic* environment. We derived the data fields, tasks, and representations to study through a combination of an existing survey [18], our own survey, and our own experience. We devised five tasks about three commonly used data fields (distance, pace and heart rate), and for each we designed one TEXT and one VISUALIZATION representation that we implemented in two data pages (DP1 and DP2, see Figure 1). Our results show that runners i) complete the tasks much faster with VISUALIZATION, ii) prefer VISUALIZATION to TEXT for those tasks, and iii) would like to have access to such visualizations while running.

2 SURVEY OF RUNNER’S DATA NEEDS

The first step in this research was to identify which data fields runners rely on while running with a smart watch, and how these data fields are being displayed. One survey [18] found that 2,218 runners that used smart watches relied on Distance (96%), Elapsed Time (90%), Speed/Pace (85.5%) and Heart Rate (68.2%). Building on this work, we designed an online survey targeting people who run at least once a week and use smart watches to track their activity. The survey was available for a month and we advertised through posters on our university campus and on social media.

SURVEY DESIGN – Participants completed a consent form then filled out a questionnaire asking about their age, gender, running experience and frequency, experience with smart watches while running, and brand of watch they use (see supplemental material).

They were then asked to indicate the data page they use for different types of running activities (base, long, tempo and progression runs, intervals, fartlek, hill repeats and sprints). They were shown data pages that correspond to their watch brand/make, and could specify their own by uploading images or describing them in text.

Next, participants were asked to rate the frequency at which they access 20 data fields while running (based on the data fields provided in existing running-specific smart watches), from Never (0%) to Sometimes (30%) to Often (70%) to Always (100%). They could also indicate data fields not included in the survey.

PARTICIPANTS – We collected 19 complete and authentic responses for analysis. Participants were aged under 20 (21%), 20–30 (47%), 30–40 (21%), 40–50 (6%) and over 50 (5%). 26% were female and 74% were male. They had been running for 1 year (16%), two years (37%) and three years and more (47%). All were running with a smart watch. They were running twice a week (37%), 3–5 times a week (53%) and more than five times a week (10%).

RESULTS – We aggregated the results for all types of running activities, because the responses did not reveal any differences.

► *How many data fields?* Two data pages were selected much more often than the others — 7 and 6 times compared to once or twice for all the other ones. These showed 3 and 5 data fields respectively, therefore we decided to display four data fields.

► *Which data fields?* Time of day, Elapsed Time, Pace (Speed), Distance, and Heart Rate were the most used data fields across all data pages and running activity types — replicating previous findings [18]. We discarded Time of Day because it is not a running-specific data field. We included Distance (present in 100% of all the submitted data pages), Pace (65%), Elapsed Time (55%), and Heart Rate (44%). These data fields are important to runners and commonly found in existing data pages; they are also of varied data types and inform different types of tasks, as we explain next.

RUNNERS' TASKS – We combined the survey results with the last author's extensive experience with running with a smart watch to iteratively define five meaningful tasks about the selected data fields.

Elapsed Time and Distance are fundamental data fields that only increase over time. Common runner questions associated to this data type include knowing the actual number and getting a sense of completion (how far in their run they are). Given the similarity of these data fields, we only formulated the following task instances for Distance (that would similarly apply to Elapsed Time).

► *T_{Dist}*: Have you finished the first quarter of your run? Are you closer to the end or to the beginning of your run? Do you have less than a quarter of your run left? These three instances ask runners to provide a binary response based on their estimation of the proportion of their run they have completed.

Pace (or speed) is important in workouts and races. Target pace is often dynamic, e.g., when performing high-intensity interval training. Common runner questions about Pace include knowing their current pace, knowing whether they are on pace or they should run slower or faster based on their activity goals, and by how much.

► *T_{Pace?}*: Is your pace above or below the target pace? This task asks runners to make a binary decision.

► *T_{PaceAdj}*: Your pace is below/above the target. How much do you need to adjust it? The two variations of this task ask runners to estimate the extent to which they need to adjust their pace.

Heart Rate is a measure of effort, and in running, it is often utilized with respect to heart rate zones. For example, one might define a heart-rate zone for an 'easy pace' between 120 and 130 bpm and attempt to stay within that zone during an activity.

► *T_{HR?}*: Is your heart rate in the target range? This task asks runners to make a binary decision.

► *T_{HRadj}*: Your Heart Rate is below/above target, how much do you need to adjust it? The two variations of this task ask runners to estimate the extent to which they need to adjust their heart rate.

These five tasks capture different data types and kinds of responses: binary decision based on proportion estimation (*T_{Dist}*), a target value (*T_{Pace?}*), or a target range (*T_{HR?}*); and estimation of deviation from a target value (*T_{PaceAdj}*) or range (*T_{HRadj}*).

3 DESIGN OF RUNNING DATA PAGES

We designed data pages on a Fossil Gen6 with a round screen resolution of 416 × 416 pixels and an AMOLED display, compatible with the Wear OS. To include the four data fields, we split the watch screen into 4 parts — a standard layout in running data pages

(see Figure 1). We designed two data pages (DP1 and DP2) that would allow us to study runners' ability to complete tasks with either TEXT or VISUALIZATION. Our design choices are grounded in the survey results, the last author's running experience, several rounds of design ideation and discussion among the researchers, and a review of how the selected data fields are represented on data pages (e.g., progress bars for goal-oriented metrics, speedometers for interval training and line charts for elevation profile).

► *Elapsed Time*: was always represented with TEXT (the study did not include tasks related to Elapsed Time).

► *Distance*: TEXT shows the number of kilometers already run over the number of kilometers to run. VISUALIZATION is a rectilinear progress bar that gets filled based on distance already run from 0 to the target distance — an effective part-to-whole representation [2].

► *Pace*: TEXT shows the current pace with a large font and the target pace with a small font. VISUALIZATION is a dual bar chart with a baseline centered at the target pace (the gray vertical line). A rectangular indicator shows the current pace. It is placed on the left bar if the current pace is lower than the target pace at a distance proportional to the pace difference, and on the right bar if it is higher. It also includes a small label that shows the exact pace value.

► *Heart Rate*: TEXT shows the current heart rate value with a large font and the target range with a small font. VISUALIZATION is a speedometer with three zones — a common representation on running data pages. The green zone is the target zone within the desired range. The yellow and red zones correspond to heart rate values below and above the target range. An arrow shows where the current heart rate value falls, and a label shows the exact value.

In summary, DP1 shows Heart Rate with TEXT and Pace and Distance with VISUALIZATION, and DP2 shows Pace and Distance with TEXT and Heart Rate with VISUALIZATION. Each data page includes a mix of TEXT and VISUALIZATION representations (instead of one with only TEXT and one with only VISUALIZATION) to limit the risk that the look of a data page influences the results.

4 STUDY DESIGN AND METHODOLOGY

HYPOTHESES – We formulated 5 hypotheses (one for each task) regarding completion time. We did not formulate hypotheses regarding accuracy because our pilot studies showed that accuracy was near perfect for all tasks with both representations.

We hypothesized that all tasks would be completed faster with VISUALIZATION than with TEXT, for the following reasons. For *T_{Dist}*, VISUALIZATION supports visual estimation of proportions, while TEXT requires mental calculation; this is also supported by previous work that found that people are faster with rectilinear part-to-whole visualizations than with text [2]. For *T_{Pace?}*, VISUALIZATION requires reading the position of the marker compared to the baseline, while TEXT requires reading and comparing two number symbols. For *T_{PaceAdj}*, VISUALIZATION requires estimating length (the vertical distance between the marker and the baseline), while TEXT requires reading and estimating the proportion between two numbers. For *T_{HR?}*, VISUALIZATION requires identifying the zone the arrow points to, while TEXT requires reading three numbers and determining if one value is both higher than the minimum value and smaller than the maximum value. For *T_{HRadj}*, VISUALIZATION requires estimating the angle between the arrow and a virtual vertical line, while TEXT requires reading three numbers and estimating the proportion between one value and a range.

TASKS AND MEASUREMENTS – Participants performed 26 tasks in total (13 tasks with each data page).

They completed each binary task three times (*T_{Dist}*, *T_{Pace?}* and *T_{HR?}*) by answering 'Yes' or 'No', 'beginning' or 'end', or 'Above' or 'Below'. They completed each estimation task twice (*T_{PaceAdj}* and *T_{HRadj}*) by answering 'A little', 'Somewhat', or 'A lot'. To avoid problems of subjective interpretation of those terms, no task

repetition had 'Somewhat' for correct answer (but participants did not know this), therefore the only two repetitions.

To ensure consistency between participants, we generated a credible dataset that was used by all participants — this means the tasks were not based on participants' actual running data.

Participants were asked to look at the watch only when prompted to complete a task, as follows: i) the experimenter asked the question verbally to the participant; ii) upon hearing a trigger sound at the end of the question, the participant looked at the watch; iii) the participant answered the question verbally. Completion time is the difference between the time the participant started providing the answer minus the time at which the trigger sound was played.

SETUP AND PROCEDURE – We conducted the study on a 400m oval track at the University of Victoria. Participants confirmed their eligibility and consent to participate, and filled out a pre-study questionnaire. After they were explained the experiment, participants were equipped with a Fossil Gen 6 smart watch (amoled display of screen resolution 416 × 416), and wireless airpods for audio connection with the experimenter who stood near the track. Participants were randomly assigned to one of two groups, that determined the order in which they would use DP1 and DP2.

Participants completed the 13 tasks with the first data page in a 10-minute long run around the track, with tasks performed at pre-determined times, with 20–60 seconds in between. Then, they were given a 10-minute break to recover and answer three questions for each representation (either TEXT or VISUALIZATION) for the three data fields — asking about clarity and usefulness, and whether they would use the representation if it was supported by their watch. They answered on 5-point Likert scales (1: Strongly disagree, 2: Somewhat disagree, 3: Neither agree nor disagree, 4: Somewhat agree, 5: Strongly agree). Participants then ran, completed the 13 tasks, and indicated their preferences with the second data page.

Next, for each data field, participants rated how appropriate each data page (and its representation for that data field) was, using the same 5-point Likert scale. The study ended with a semi-structured interview in which participants could talk about their experience.

PARTICIPANTS – We recruited 20 new runner participants (13M, 7F) through mailing lists, social media and posters displayed on campus. 15 were students; they were aged 19–43 (mean 24.6); were running once (1), twice (5), and more than four times (14) a week; and all ran with a smart watch. They had an undergraduate degree (15), a Masters (3), a PhD (1) or did not answer (1). They had backgrounds in Physical and Health Education (11), Computer Science and Engineering (5), Psychology (3) or did not answer (1).

DATA COLLECTION AND ANALYSIS – We analyzed completion times using Bootstrapped [21] confidence intervals (CIs) rather than p -values, following recommendations for statistical practices in HCI and visualization (e.g., [10, 12, 3, 6]). We interpret the statistical significance of the overlap of CI bars following [23]. We performed analyses with 95% CIs computed with 1000 replicates and the BCa method [21], using the mean of the repetitions as the aggregated measure for each $\{participant, task, representation\}$ triplet.

5 RESULTS

COMPLETION TIME – see Figure 2. Pairwise comparisons show that participants were significantly faster with VISUALIZATION than with TEXT for all tasks (all hypotheses are confirmed).

The effects are large for all tasks: participants were 3.2 times slower with TEXT than with VISUALIZATION for T_{Dist} , 1.8 times slower for $T_{Pace?}$, 2.9 times slower for $T_{PaceAdjust}$, 1.6 times slower for $T_{HR?}$, and 7.9 times slower for $T_{HRRadjust}$.

PREFERENCES – see Figure 3. Overall, VISUALIZATION was largely preferred for both Distance and Heart Rate, and slightly preferred for Pace. For both Distance and heart Rate and for all

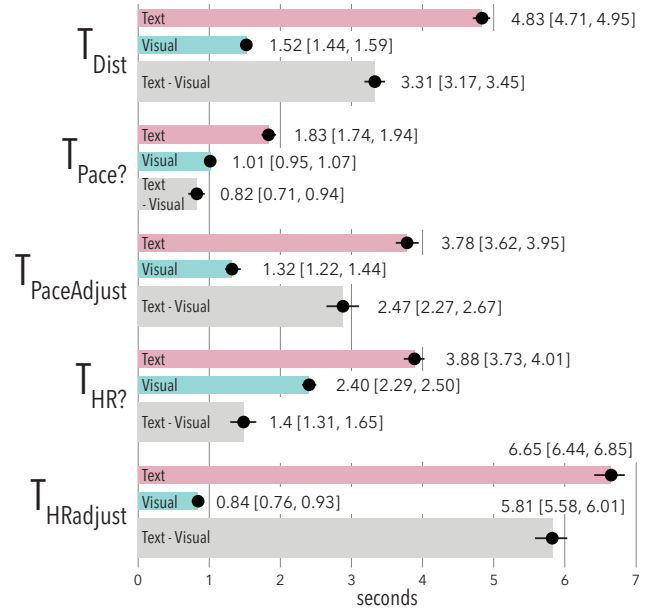


Figure 2: Bootstrapped mean completion time (in seconds) with TEXT (in pink) and VISUALIZATION (in teal) for the five tasks, along with the bootstrapped mean differences in seconds between TEXT and VISUALIZATION (in grey). Bars show 95% CI. Mean estimates and CI values are shown besides each bar.

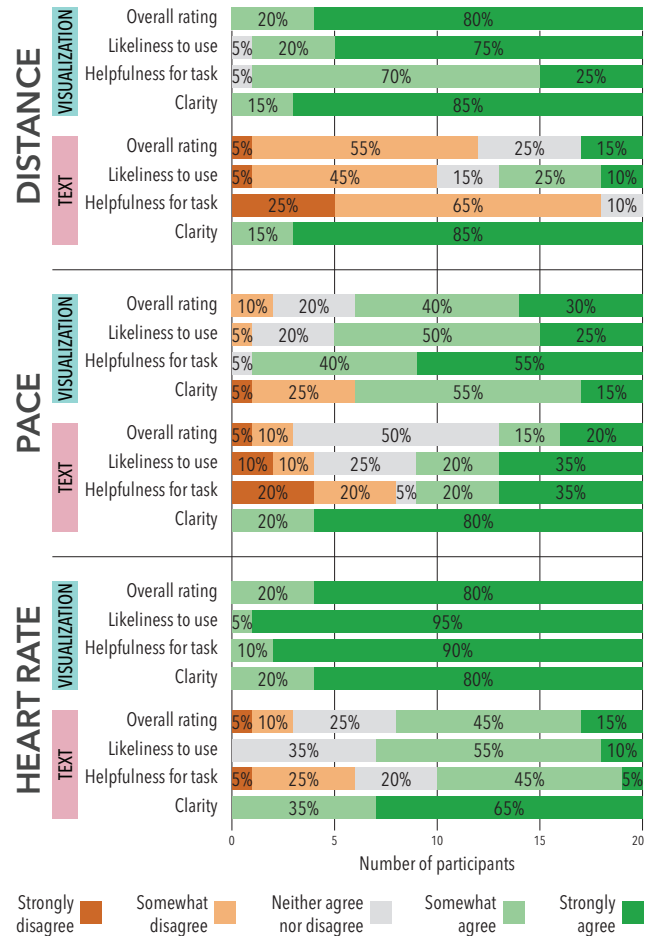


Figure 3: Subjective ratings for TEXT and VISUALIZATION.

questions, at least 95% of the participants agree that VISUALIZATION is clear and helpful for the task and that they are likely to use such VISUALIZATION in their running activities. Participants are much less positive about TEXT, especially for Distance.

The difference is less strong for Pace. In particular, although VISUALIZATION is overall preferred to TEXT, participants also found VISUALIZATION less clear than TEXT for that data field.

6 DISCUSSION

We contextualize the results with quotes from participants during the post-study interview to discuss i) the usefulness of visualizations when running, ii) how context affects runners' representational needs, and iii) methodological limitations and opportunities.

VISUALIZATIONS ARE INDEED USEFUL WHEN RUNNING – Even with simple tasks that involve only few numbers, visualization helped significantly. Participants were faster with VISUALIZATION than with TEXT for all tasks, and they were often aware of it, e.g., P1 said *“I quickly could see my progress visually, without having to check numerical values.”* Several participants said that they struggled to complete T_{Dist} with TEXT because of the cognitive load associated with performing mental math while running, e.g., P10 said *“I don't like to do any sort of math or thinking so something that shows me where I am and where I need to be is pretty handy.”*

Interestingly, although participants were more efficient with VISUALIZATION than with TEXT across the board, the subjective ratings for Pace were only slightly in favor of VISUALIZATION. Participants explained, for example, *“I prefer [the] numeric one, [it's] easier to just see the current pace”* (P14), and *“I am more comfortable with seeing a big number instead of bars and colors.”* (P3).

Overall, **visualization is more than a viable alternative to numbers for runners**. P14 said *“I always find the visual representations are often easier to understand than numbers, especially when I am running or when I am doing a physical activity.”* This stresses the importance of glanceable visualizations for some data-related tasks when running. We expect this to be even truer when runners are **under limited cognitive resources** like when performing particularly strenuous activities [9] such as high-intensity interval training or racing, as well as **under a variety of real-life running environments** like on trails or in traffic where looking at a smart watch for a few seconds can lead to fall and injury (this has happened to the last author of this paper too many times).

REPRESENTATIONAL NEEDS DEPEND ON CONTEXT AND TASK – Our results are to be put in perspective of the tasks that participants performed. Many more tasks are meaningful for runners, and numbers are likely more versatile than visualizations to accomplish a wide range of tasks. Some participants recognized this, with P19 saying for example *“Visualization were good for comparing but I prefer just number to see my speed.”* P5 commented that *“[TEXT and VISUALIZATION] are very equally good to me, if both adapt my watch, based on my running plan I can choose either or.”* This outlines the usefulness to provide different modes of data representation that runners can customize based on specific running activities. Some activities are also more data-intensive than others. For example, high-intensity interval training or hill repeats require closely monitoring multiple metrics — as these are measures of performance and success. Conversely, a base run can often be completed without the need to access data at all — as the measure of success is basically to spend ‘time on your feet’.

Achieving goals is another example that illustrates how representational needs depend on context and task. Echoing past work [29], three participants explained that they felt more motivated to achieve their goals and perform well with VISUALIZATION than with TEXT for T_{Dist} . They said *“I was having a sense of accomplishment and it was encouraging me to push myself further.”* (P5); *“the visual designs were giving me this feeling to focus on the goal.”* (P16); and

“the bar that was filling with color as I was running was giving me more motivation to fill it as soon as possible.” (P8). Future work should look specifically into **the effect of different visualization designs on goal-setting and motivation**.

METHODOLOGICAL LIMITATIONS AND OPPORTUNITIES – These study results apply to the set of tasks we tested and with the demographic of runners we recruited. We expect that different tasks would lead to different results. For example, people could be faster with TEXT than with VISUALIZATION to retrieve exact values; and also more accurate if VISUALIZATION would not provide labels of those values. In the future, we plan to establish a complete list of meaningful tasks for runners based on context (activity type, environment, social setting, etc..) so that further studies can investigate the usefulness of visualization for these.

Tasks and data needs will also greatly vary depending on runner's expertise and practice. Some run a turkey trot once a year, some are weekend warriors, some are elite athletes or professional runners; people run on tracks and roads, and in forests and mountains; they run alone or with company — human and animal alike; they run in the sun, snow or pouring rain; they run 1km, 10km, or 200km. Future work needs to study all these factors that make up the context which affects tasks and representational needs.

Conducting this study forced us to rethink and adapt the way we typically conduct studies in HCI and visualization. While it has a long history of conducting quantitative in-lab controlled experiments, the community seldom conducts such studies in realistic environments — likely because it is much more difficult. As a field, we do not have the same infrastructure, knowledge and history of practice about such studies, and it proved hard to find information about appropriate methods to use in this research. This is at the same time a challenge, and an opportunity to research and implement new methodological frameworks.

In our study we decided to use a synthetic dataset to maximize task difficulty consistency across participants; but that is at the cost of less realistic data that might have distracted some participants, who might have realized that the data was not accurate. Another decision we made was to ask questions and collect responses verbally. This comes with technological challenges (how to send the prompt, how to measure the time it takes to respond, how to collect the participant data) and is imperfect (Bluetooth transmission likely introduced some noise in the response times collected). All these factors are threats to external and internal validity and raise questions around the difficulty to fully automate this type of study, how to administer tasks, and how to collect participant data.

7 CONCLUSIONS

We conducted a study in which 20 runners completed running-related tasks on an outdoor track using both text and visualizations. We found that runners perform those tasks much faster with visualizations than with text, prefer visualizations to text, and would use such visualizations while running.

Running is an untapped context for visualization: runners are data-nerds, they wear data trackers that capture dozens of data fields of many types, they perform a variety of running activities that come with different data needs and tasks, and they run in a variety of environments with many varying factors (physical, psychological, meteorological, social, geographical, etc..). Studying how runners (and other outdoor enthusiasts) use visualizations requires going beyond the ‘easy’ treadmill setup and actually considering all those factors in study design. What we learn about how runners use visualization in these realistic environments and scenarios, with runners being under stress, duress, fatigue, limited cognition and abilities, and where looking at a visualization is not the primary task, likely applies to other environments where visualizations might be used with similar limited attentional capacity.

ACKNOWLEDGMENTS

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SUPPLEMENTAL MATERIAL

- `study-protocol.pdf`: includes the protocol, pre-study questionnaire, exact tasks and data page stimuli, and post-task questionnaires used in the study.
- `survey.pdf`: the survey used for the preliminary study.

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